John Pumayalli

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EDUCATION

Stanford University Expected June 2027
BS Symbolic Systems — AI & Robotics Concentration · *GPA: 3.61* Stanford, CA
Coursework: Machine Learning, Algorithms, Mathematical Foundations, Probability, Computer Organization, Mathematical Logic, Minds & Machines, Philosophy of AI, Semantics & Pragmatics

Telluride High School June 2023
GPA: 4.34 · *Magna Cum Laude* · *Ronald Reagan Leadership Medal* Telluride, CO

EXPERIENCE

OrgOS March 2026 – Present
Software Engineer Intern Stanford, CA
- Built a production Python pipeline ingesting Slack data from NeonDB and using Gemini 2.5 Flash (OpenRouter) to infer org structure — actors, roles, seniority scores, team groupings — across thousands of messages
- Architected GCP deployment with Secret Manager, CI/CD via GitHub Actions, and Terraform-managed infrastructure; validated all LLM outputs as structured JSON before NeonDB write

Stanford Sierra Camp June – Sep 2025
Camp Counselor South Lake Tahoe, CA
- Designed and led daily programming for 300+ weekly guests over a 13-week season; built relationships across a Stanford alumni network

Flourish June – Sep 2022
Front End Developer Dartmouth, NH
- Built patient-facing interface for a chronic-condition health app using React Native and Node.js; constructed MongoDB backend in coordination with the UI/UX team

Colorado School of Mines June – Sep 2021
Research Assistant · *Dr. Chuan Yue* Golden, CO
- Reverse-engineered Grammarly to identify privacy vulnerabilities and cybersecurity issues in the browser extension

PROJECTS

Beyond Pip Counting: Interpretable ML for Catan Placement Stanford CS229
Python · XGBoost · SHAP · scikit-learn · k-means
- Lifted win-prediction AUC from 0.54 (pip-only baseline) to 0.73 using 52 engineered features across resource scarcity, turn order, opponent competition, and strategic engines — trained on ~100K simulated games
- Applied SHAP to attribute XGBoost predictions; distilled model into a shallow decision tree where 8 human-readable rules recovered 62% of its edge over pip counting

Colonist.io 1v1 Post-Game Analyzer Personal Project - https://catan-bot-two.vercel.app/
JavaScript · Python · MessagePack · Chrome Extension (MV3) · Catanatron
- Reverse-engineered Colonist.io's real-time WebSocket/MessagePack protocol and built a Chrome MV3 extension to record games, plus a translator mapping its 54-vertex / 72-edge board onto the Catanatron rules engine.
- Scored every player decision by win probability using seeded Monte-Carlo rollouts, flagging statistically significant blunders and surfacing the engine's preferred move.

LEADERSHIP & ACTIVITIES

Sigma Phi Epsilon — Social Chair & Head of Member Development Sep 2024 – Present

Mestiza Qoyllacha — Member & Performer Sep 2012 – Present
Annual collaborator with an Indigenous dance collective in the Sacred Valley of Peru; developed cross-cultural communication and teamwork.

SKILLS

Languages: Python, C++, JavaScript, Java, Swift, HTML/CSS, SQL · Spanish (native), English (native), Italian (intermediate)
ML / AI: XGBoost, scikit-learn, SHAP, LLM APIs, OpenRouter, Gemini 2.5, logistic regression, k-means
Infrastructure: GCP, GitHub Actions, Terraform, PostgreSQL, NeonDB, MongoDB
Frameworks: React, React Native, Node.js, Figma, CAD